



## Qualification Pack

# Seaweed Grower and Processor

QP Code: CUTM/AGR/Q0202

Version: 1.0

NSQF Level: 4

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## Qualification Pack

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## Qualification Pack

### CUTM/AGR/Q0202: Seaweed Grower and Processor

#### Brief Job Description

The Individual is responsible to understand about Seaweed Grower and Processor, check water quality, nutrition, and health benefits and indirectly evaluate the quality parameters of fish and fishery products.

#### Personal Attributes

A Seaweed Grower and Processor should be environmentally responsible, physically fit, and attentive to detail. They must have a strong understanding of marine ecosystems and sustainable practices. Patience, precision, and a commitment to hygiene and quality are essential for post-harvest handling and processing. Creativity in developing value-added products, along with problem-solving skills, adaptability to coastal conditions, and dedication to eco-friendly methods, are key traits for success in this role.

#### Applicable National Occupational Standards (NOS)

##### Compulsory NOS:

1. [CUTM/AGR/N0201: Seaweed Ecology and its environmental importance](#)
2. [CUTM/AGR/N0202: Post-Harvest handling of Seaweed](#)
3. [CUTM/AGR/N0203: Value added seaweed product and preservation](#)
4. [CUTM/AGR/N0204: Quality Assurance in seaweed processing](#)
5. [CUTM/AGR/N0205: Minimizing the use of synthetic pesticides and fertilizers and practicing the use of organic and sustainable principles](#)
6. [CUTM/AGR/N0213: Implementation of Seaweed safety standards](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

#### Qualification Pack (QP) Parameters

|                   |                 |
|-------------------|-----------------|
| <b>Sector</b>     | Agriculture     |
| <b>Sub-Sector</b> | Agri Harvesting |
| <b>Occupation</b> | Post-Harvest    |
| <b>Country</b>    | India           |



## Qualification Pack

|                                                           |                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NSQF Level</b>                                         | 4                                                                                                                                                                                                                                                                                     |
| <b>Credits</b>                                            | 17                                                                                                                                                                                                                                                                                    |
| <b>Aligned to NCO/ISCO/ISIC Code</b>                      | NSQF/NCrF                                                                                                                                                                                                                                                                             |
| <b>Minimum Educational Qualification &amp; Experience</b> | 12th grade Pass with 1 Year of experience relevant experience<br>OR<br>Previous relevant Qualification of NSQF Level (3.5) with 1.5 years of experience relevant experience<br>OR<br>Previous relevant Qualification of NSQF Level (3) with 3 Years of experience relevant experience |
| <b>Minimum Level of Education for Training in School</b>  |                                                                                                                                                                                                                                                                                       |
| <b>Pre-Requisite License or Training</b>                  | NA                                                                                                                                                                                                                                                                                    |
| <b>Minimum Job Entry Age</b>                              | 19 Years                                                                                                                                                                                                                                                                              |
| <b>Last Reviewed On</b>                                   | NA                                                                                                                                                                                                                                                                                    |
| <b>Next Review Date</b>                                   | 21/10/2027                                                                                                                                                                                                                                                                            |
| <b>NSQC Approval Date</b>                                 | 21/10/2024                                                                                                                                                                                                                                                                            |
| <b>Version</b>                                            | 1.0                                                                                                                                                                                                                                                                                   |
| <b>Reference code on NQR</b>                              | QG-04-AG-03352-2024-V1-CUTM                                                                                                                                                                                                                                                           |
| <b>NQR Version</b>                                        | 1.0                                                                                                                                                                                                                                                                                   |

## Qualification Pack

### CUTM/AGR/N0201: Seaweed Ecology and its environmental importance

#### Description

This NOS covers the ecological importance of seaweed, its role in marine ecosystems, and sustainable cultivation methods. It includes species identification, environmental monitoring, and conservation strategies to promote biodiversity.

#### Scope

The scope covers the following :

- The scope includes identifying seaweed species, monitoring water quality, understanding ecological roles, and implementing sustainable cultivation methods to enhance biodiversity while ensuring environmental protection.

#### Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Various types of seaweed, their economic and environmental importance, and the global seaweed industry.
- PC2.** Identification and classification of seaweed species through field visits.
- PC3.** Understanding seaweed anatomy and physiology.
- PC4.** Growth conditions: light, temperature, salinity, and nutrients.
- PC5.** Seaweed samples and testing water quality parameters.
- PC6.** Role of seaweed in marine ecosystems.

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** different seaweed species and their ecological significance.
- KU2.** seaweed's role in carbon sequestration and climate change mitigation.
- KU3.** about marine biodiversity and seaweed's contribution to habitat formation.
- KU4.** nutrient cycling and water quality improvement by seaweed.
- KU5.** environmental parameters affecting seaweed growth.
- KU6.** sustainable seaweed cultivation methods.

#### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Conduct field surveys to analyze seaweed populations.
- GS2.** Perform water quality tests for salinity, temperature, and nutrients.
- GS3.** Identify and classify various seaweed species.



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- GS4.** Apply sustainable seaweed farming techniques.
- GS5.** Document and interpret seaweed research data.
- GS6.** Apply GIS and remote sensing tools for mapping seaweed habitats.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                    | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                                                     | 30           | 70              | -             | -          |
| <b>PC1.</b> Various types of seaweed, their economic and environmental importance, and the global seaweed industry. | 5            | 10              | -             | -          |
| <b>PC2.</b> Identification and classification of seaweed species through field visits.                              | 5            | 20              | -             | -          |
| <b>PC3.</b> Understanding seaweed anatomy and physiology.                                                           | 5            | 10              | -             | -          |
| <b>PC4.</b> Growth conditions: light, temperature, salinity, and nutrients.                                         | 5            | 10              | -             | -          |
| <b>PC5.</b> Seaweed samples and testing water quality parameters.                                                   | 5            | 10              | -             | -          |
| <b>PC6.</b> Role of seaweed in marine ecosystems.                                                                   | 5            | 10              | -             | -          |
| <b>NOS Total</b>                                                                                                    | <b>30</b>    | <b>70</b>       | -             | -          |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                  |
|----------------------------|--------------------------------------------------|
| <b>NOS Code</b>            | CUTM/AGR/N0201                                   |
| <b>NOS Name</b>            | Seaweed Ecology and its environmental importance |
| <b>Sector</b>              | Agriculture                                      |
| <b>Sub-Sector</b>          |                                                  |
| <b>Occupation</b>          | Post-Harvest                                     |
| <b>NSQF Level</b>          | 4                                                |
| <b>Credits</b>             | 2                                                |
| <b>Version</b>             | 1.0                                              |
| <b>Last Reviewed Date</b>  | 21/10/2024                                       |
| <b>Next Review Date</b>    | 21/10/2027                                       |
| <b>NSQC Clearance Date</b> | 21/10/2024                                       |

## Qualification Pack

### CUTM/AGR/N0202: Post-Harvest handling of Seaweed

#### Description

This NOS focuses on post-harvest handling of seaweed, covering cleaning, sorting, drying, preservation, and storage to maintain product quality and extend shelf life while ensuring food safety compliance.

#### Scope

The scope covers the following :

- The scope includes best practices for seaweed harvesting, handling, processing, and packaging, ensuring minimal spoilage and adherence to safety and quality standards for market distribution.

#### Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Cleaning of seaweed as they may contain many impurities such as sand and debris.
- PC2.** It should be sorted and graded according to their size, color, species and quality.
- PC3.** One of the pivotal factors in seaweed post harvesting is drying process.
- PC4.** Proper packaging is important for maintaining quality and freshness.
- PC5.** Gain insights into the global seaweed market, its economic potential and current trends in supply and demand.
- PC6.** Develop entrepreneurial skills.
- PC7.** Quality control measures are implemented to meet the regulatory standards and customer requirements.

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** best practices for harvesting seaweed to maintain quality.
- KU2.** techniques for cleaning and sorting seaweed post-harvest.
- KU3.** proper drying and preservation methods.
- KU4.** storage conditions to prevent spoilage.
- KU5.** logistics and transportation of seaweed products.
- KU6.** quality grading and sorting techniques.

#### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Perform proper post-harvest seaweed cleaning techniques.
- GS2.** Apply sorting and grading methods to ensure quality.



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- GS3.** Operate drying and preservation equipment.
- GS4.** Implement HACCP protocols for safe handling.
- GS5.** Work with teams to process seaweed efficiently.
- GS6.** Develop problem-solving skills for post-harvest challenges.
- GS7.** Handle packaging for long-term storage.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                          | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                                                           | 30           | 40              | -             | -          |
| <b>PC1.</b> Cleaning of seaweed as they may contain many impurities such as sand and debris.                              | 4            | 6               | -             | -          |
| <b>PC2.</b> It should be sorted and graded according to their size, color, species and quality.                           | 4            | 5               | -             | -          |
| <b>PC3.</b> One of the pivotal factors in seaweed post harvesting is drying process.                                      | 5            | 6               | -             | -          |
| <b>PC4.</b> Proper packaging is important for maintaining quality and freshness.                                          | 4            | 5               | -             | -          |
| <b>PC5.</b> Gain insights into the global seaweed market, its economic potential and current trends in supply and demand. | 5            | 6               | -             | -          |
| <b>PC6.</b> Develop entrepreneurial skills.                                                                               | 4            | 6               | -             | -          |
| <b>PC7.</b> Quality control measures are implemented to meet the regulatory standards and customer requirements.          | 4            | 6               | -             | -          |
| <b>NOS Total</b>                                                                                                          | <b>30</b>    | <b>40</b>       | <b>-</b>      | <b>-</b>   |



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### National Occupational Standards (NOS) Parameters

|                            |                                  |
|----------------------------|----------------------------------|
| <b>NOS Code</b>            | CUTM/AGR/N0202                   |
| <b>NOS Name</b>            | Post-Harvest handling of Seaweed |
| <b>Sector</b>              | Agriculture                      |
| <b>Sub-Sector</b>          |                                  |
| <b>Occupation</b>          | Post-Harvest                     |
| <b>NSQF Level</b>          | 4                                |
| <b>Credits</b>             | 4                                |
| <b>Version</b>             | 1.0                              |
| <b>Last Reviewed Date</b>  | 21/10/2024                       |
| <b>Next Review Date</b>    | 21/10/2027                       |
| <b>NSQF Clearance Date</b> | 21/10/2024                       |

## Qualification Pack

### CUTM/AGR/N0203: Value added seaweed product and preservation

#### Description

This NOS covers the processing and preservation of seaweed into value-added products such as food, pharmaceuticals, and cosmetics while ensuring safety and quality compliance.

#### Scope

The scope covers the following :

- The scope includes extraction, preservation, packaging, and marketing of value-added seaweed products while maintaining high standards of quality and sustainability.

#### Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Extraction process for obtaining secondary metabolites such as proteins, polyphenols and polysaccharides.
- PC2.** Extract from seaweeds can be used as an ingredient in food products.
- PC3.** The extracts are commonly used in skincare, hair care and personal care products.
- PC4.** Seaweed-based bioplastics offer a sustainable alternative to petroleum-based plastics for packaging and disposable products.
- PC5.** Seaweed is used as a nutritious feed ingredient for livestock, aquaculture, and pet food due to its high protein, mineral, and vitamin content.
- PC6.** Seaweed supplements for pets are also available to promote overall health and vitality.
- PC7.** Preservation methods such as drying, freezing, pickling, fermentation, and irradiation are employed to extend the shelf life of seaweed products while retaining their quality, flavor, and nutritional value.
- PC8.** Seaweed extracts are utilized as organic fertilizers and soil conditioners in agricultural practices.

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** methods for producing value-added seaweed products.
- KU2.** preservation techniques like drying, freezing, and fermentation.
- KU3.** bioactive compounds extracted from seaweed for medicinal uses.
- KU4.** about seaweed-based food additives and ingredients.
- KU5.** sustainable packaging for processed seaweed.
- KU6.** regulatory guidelines for processed seaweed products.

#### Generic Skills (GS)



## Qualification Pack

User/individual on the job needs to know how to:

- GS1.** Develop skills for processing seaweed into consumable products.
- GS2.** Apply preservation techniques to maintain nutritional value.
- GS3.** Operate machinery for drying and extraction.
- GS4.** Ensure compliance with safety standards in processing.
- GS5.** Document product formulations and modifications.
- GS6.** Identify quality control measures for processed products.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                           | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                                                                                                                                                            | 40           | 60              | -             | -          |
| <b>PC1.</b> Extraction process for obtaining secondary metabolites such as proteins, polyphenols and polysaccharides.                                                                                                      | 5            | 7               | -             | -          |
| <b>PC2.</b> Extract from seaweeds can be used as an ingredient in food products.                                                                                                                                           | 5            | 7               | -             | -          |
| <b>PC3.</b> The extracts are commonly used in skincare, hair care and personal care products.                                                                                                                              | 5            | 7               | -             | -          |
| <b>PC4.</b> Seaweed-based bioplastics offer a sustainable alternative to petroleum-based plastics for packaging and disposable products.                                                                                   | 5            | 7               | -             | -          |
| <b>PC5.</b> Seaweed is used as a nutritious feed ingredient for livestock, aquaculture, and pet food due to its high protein, mineral, and vitamin content.                                                                | 5            | 7               | -             | -          |
| <b>PC6.</b> Seaweed supplements for pets are also available to promote overall health and vitality.                                                                                                                        | 5            | 8               | -             | -          |
| <b>PC7.</b> Preservation methods such as drying, freezing, pickling, fermentation, and irradiation are employed to extend the shelf life of seaweed products while retaining their quality, flavor, and nutritional value. | 5            | 8               | -             | -          |
| <b>PC8.</b> Seaweed extracts are utilized as organic fertilizers and soil conditioners in agricultural practices.                                                                                                          | 5            | 9               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                                                           | <b>40</b>    | <b>60</b>       | <b>-</b>      | <b>-</b>   |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                              |
|----------------------------|----------------------------------------------|
| <b>NOS Code</b>            | CUTM/AGR/N0203                               |
| <b>NOS Name</b>            | Value added seaweed product and preservation |
| <b>Sector</b>              | Agriculture                                  |
| <b>Sub-Sector</b>          |                                              |
| <b>Occupation</b>          | Post-Harvest                                 |
| <b>NSQF Level</b>          | 4                                            |
| <b>Credits</b>             | 3                                            |
| <b>Version</b>             | 1.0                                          |
| <b>Last Reviewed Date</b>  | 21/10/2024                                   |
| <b>Next Review Date</b>    | 21/10/2027                                   |
| <b>NSQC Clearance Date</b> | 21/10/2024                                   |

## Qualification Pack

### CUTM/AGR/N0204: Quality Assurance in seaweed processing

#### Description

This NOS focuses on ensuring the quality and safety of processed seaweed products through testing, regulatory compliance, and adherence to global food safety standards.

#### Scope

The scope covers the following :

- The scope includes conducting quality checks, maintaining hygiene standards, ensuring safety compliance, and implementing quality control measures in seaweed processing.

#### Elements and Performance Criteria

##### *Understanding the different quality parameters associated with seaweed harvesting*

To be competent, the user/individual on the job must be able to:

- PC1.** Sustainable harvesting practices and regulations to ensure long-term health of the seaweed.
- PC2.** Effective methods for cleaning, sorting, and storing harvested seaweed to maintain quality and prevent spoilage.

##### *Knowledge on various methods of assessment and maintenance of quality*

To be competent, the user/individual on the job must be able to:

- PC3.** Exploring various drying techniques like sun dry, oven dry and their impact on seaweed quality, nutritional value, and shelf life.
- PC4.** Learning about different methods for extracting valuable compounds from seaweed, such as alginates, carrageenan, agar, and fucoxanthin, for various research and commercial applications.
- PC5.** Discover processes for converting seaweed into products such as food products, cosmetics, fertilizers, biofuels, and other marketable goods.
- PC6.** Understanding the importance of quality control measures in the entire processing method and gaining knowledge on relevant regulations for food safety and product labeling.
- PC7.** Regular visit to commercial seaweed processing plants can provide valuable real-world insights into the industrial and commercial sector.

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** quality parameters for seaweed processing.
- KU2.** microbial testing for seaweed products.
- KU3.** global certification standards for seaweed processing.
- KU4.** techniques for maintaining nutritional integrity.
- KU5.** packaging quality control measures.
- KU6.** environmental safety in seaweed processing plants.



## Qualification Pack

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Conduct quality testing of seaweed products.
- GS2.** Implement safety protocols in processing facilities.
- GS3.** Operate lab equipment for microbial and chemical analysis.
- GS4.** Develop SOPs for quality assurance.
- GS5.** Train teams on hygiene and safety compliance.
- GS6.** Work collaboratively with suppliers and regulatory bodies.
- GS7.** Optimize processing to reduce contamination risks.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                   | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Understanding the different quality parameters associated with seaweed harvesting</i>                                                                                                           | <b>12</b>    | <b>17</b>       | -             | -          |
| <b>PC1.</b> Sustainable harvesting practices and regulations to ensure long-term health of the seaweed.                                                                                            | 6            | 8               | -             | -          |
| <b>PC2.</b> Effective methods for cleaning, sorting, and storing harvested seaweed to maintain quality and prevent spoilage.                                                                       | 6            | 9               | -             | -          |
| <i>Knowledge on various methods of assessment and maintenance of quality</i>                                                                                                                       | <b>28</b>    | <b>43</b>       | -             | -          |
| <b>PC3.</b> Exploring various drying techniques like sun dry, oven dry and their impact on seaweed quality, nutritional value, and shelf life.                                                     | 6            | 8               | -             | -          |
| <b>PC4.</b> Learning about different methods for extracting valuable compounds from seaweed, such as alginates, carrageenan, agar, and fucoidan, for various research and commercial applications. | 6            | 9               | -             | -          |
| <b>PC5.</b> Discover processes for converting seaweed into products such as food products, cosmetics, fertilizers, biofuels, and other marketable goods.                                           | 5            | 9               | -             | -          |
| <b>PC6.</b> Understanding the importance of quality control measures in the entire processing method and gaining knowledge on relevant regulations for food safety and product labeling.           | 5            | 8               | -             | -          |
| <b>PC7.</b> Regular visit to commercial seaweed processing plants can provide valuable real-world insights into the industrial and commercial sector.                                              | 6            | 9               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                                   | <b>40</b>    | <b>60</b>       | -             | -          |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                         |
|----------------------------|-----------------------------------------|
| <b>NOS Code</b>            | CUTM/AGR/N0204                          |
| <b>NOS Name</b>            | Quality Assurance in seaweed processing |
| <b>Sector</b>              | Agriculture                             |
| <b>Sub-Sector</b>          |                                         |
| <b>Occupation</b>          | Post-Harvest                            |
| <b>NSQF Level</b>          | 4                                       |
| <b>Credits</b>             | 2                                       |
| <b>Version</b>             | 1.0                                     |
| <b>Last Reviewed Date</b>  | 21/10/2024                              |
| <b>Next Review Date</b>    | 21/10/2027                              |
| <b>NSQC Clearance Date</b> | 21/10/2024                              |

## Qualification Pack

# CUTM/AGR/N0205: Minimizing the use of synthetic pesticides and fertilizers and practicing the use of organic and sustainable principles

## Description

This NOS emphasizes reducing synthetic pesticide use and adopting organic, sustainable seaweed farming practices for environmental conservation and healthier product yields.

## Scope

The scope covers the following :

- The scope includes implementing organic farming methods, minimizing synthetic inputs, ensuring soil and water conservation, and maintaining sustainable seaweed cultivation practices.

## Elements and Performance Criteria

### *Importance of sustainability in seaweed industry*

To be competent, the user/individual on the job must be able to:

- PC1.** The process of seaweed cultivation is an ecofriendly as compared to some field crops. Seaweed farming that employs sustainable methods contributes to the preservation of biodiversity, marine ecosystems, and the health of coastal regions.
- PC2.** Seaweeds play a vital role in carbon sequestration by absorbing CO<sub>2</sub>. Large-scale cultivation of seaweed helps in mitigating climatic change
- PC3.** Sustainable seaweed cultivation can be a source for reliable and resilient food.
- PC4.** Seaweeds contain photoactive compounds which has applications in various industries such as pharmaceuticals and biotechnology.

### *Environmental impact assessment*

To be competent, the user/individual on the job must be able to:

- PC5.** Evaluating the impact of seaweed cultivation on marine biodiversity which includes assessing of effect flora and fauna of local habitat. Examining the water quality, nutrient levels and O<sub>2</sub> concentrations and their effects.
- PC6.** Assessing any physical changes to the seabed or coastal regions which results from cultivation. The overall environmental can be affected by changes in geological, transparency, and structure of the habitat which needs to be studied.
- PC7.** Evaluating the risk of cultivation by introducing invasive species, by ensuring the cultivation does not cause growth of any non-native seaweed which may create an imbalance in the ecosystem.
- PC8.** Evaluating the capacity for nutrient loading in the water during the cultivation. As excessive nutrient is released during cultivation or processing may lead to eutrophication.
- PC9.** Evaluating the various technique used for pest and disease management. Chemical used and other methods should undergo an assessment on non-targeted species before the use.
- PC10.** Seaweed cultivation will lead to growth of indigenous species rather than invasive
- PC11.** Seaweed can act as a source of bioenergy.

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- PC12.** Sequestering of CO<sub>2</sub> through seaweed is deliberately done which plays a pivotal role in mitigating the climatic change.
- PC13.** The process of seaweed cultivation involves less or no use of fertilizers and pesticides which is ecofriendly than the conventional methods

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** about organic seaweed farming techniques.
- KU2.** the harmful effects of synthetic pesticides.
- KU3.** alternative pest management strategies.
- KU4.** about natural bio-fertilizers for seaweed.
- KU5.** sustainable farming principles.
- KU6.** about soil and water conservation methods.
- KU7.** government policies promoting organic farming.

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Apply organic cultivation techniques.
- GS2.** Monitor seaweed farms for pest infestations.
- GS3.** Identify organic fertilizers for optimal growth.
- GS4.** Maintain records for organic certification.
- GS5.** Conduct risk assessments for synthetic pesticide use.
- GS6.** Develop solutions for minimizing chemical inputs.
- GS7.** Implement best practices for ecological sustainability.
- GS8.** Train other farmers on organic farming techniques.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                          | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Importance of sustainability in seaweed industry</i>                                                                                                                                                                                                   | <b>13</b>    | <b>14</b>       | -             | <b>4</b>   |
| <b>PC1.</b> The process of seaweed cultivation is an ecofriendly as compared to some field crops. Seaweed farming that employs sustainable methods contributes to the preservation of biodiversity, marine ecosystems, and the health of coastal regions. | 2            | 4               | -             | 1          |
| <b>PC2.</b> Seaweeds play a vital role in carbon sequestration by absorbing CO <sub>2</sub> . Large-scale cultivation of seaweed helps in mitigating climatic change                                                                                      | 5            | 4               | -             | 1          |
| <b>PC3.</b> Sustainable seaweed cultivation can be a source for reliable and resilient food.                                                                                                                                                              | 3            | 3               | -             | 1          |
| <b>PC4.</b> Seaweeds contain photoactive compounds which has applications in various industries such as pharmaceuticals and biotechnology.                                                                                                                | 3            | 3               | -             | 1          |
| <i>Environmental impact assessment</i>                                                                                                                                                                                                                    | <b>17</b>    | <b>26</b>       | -             | <b>26</b>  |
| <b>PC5.</b> Evaluating the impact of seaweed cultivation on marine biodiversity which includes assessing of effect flora and fauna of local habitat. Examining the water quality, nutrient levels and O <sub>2</sub> concentrations and their effects.    | 4            | 4               | -             | 4          |
| <b>PC6.</b> Assessing any physical changes to the seabed or coastal regions which results from cultivation. The overall environmental can be affected by changes in geological, transparency, and structure of the habitat which needs to be studied.     | 4            | 4               | -             | 2          |
| <b>PC7.</b> Evaluating the risk of cultivation by introducing invasive species, by ensuring the cultivation does not cause growth of any non-native seaweed which may create an imbalance in the ecosystem.                                               | 1            | 3               | -             | 2          |
| <b>PC8.</b> Evaluating the capacity for nutrient loading in the water during the cultivation. As excessive nutrient is released during cultivation or processing may lead to eutrophication.                                                              | 1            | 3               | -             | 3          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                        | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC9.</b> Evaluating the various technique used for pest and disease management. Chemical used and other methods should undergo an assessment on non-targeted species before the use. | 1            | 3               | -             | 3          |
| <b>PC10.</b> Seaweed cultivation will lead to growth of indigenous species rather than invasive                                                                                         | 2            | 3               | -             | 3          |
| <b>PC11.</b> Seaweed can act as a source of bioenergy.                                                                                                                                  | 1            | -               | -             | 3          |
| <b>PC12.</b> Sequestering of CO <sub>2</sub> through seaweed is deliberately done which plays a pivotal role in mitigating the climatic change.                                         | 1            | 3               | -             | 3          |
| <b>PC13.</b> The process of seaweed cultivation of involves less or no use of fertilizers and pesticides which is ecofriendly than the conventional methods                             | 2            | 3               | -             | 3          |
| <b>NOS Total</b>                                                                                                                                                                        | <b>30</b>    | <b>40</b>       | <b>-</b>      | <b>30</b>  |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                                                                         |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>NOS Code</b>            | CUTM/AGR/N0205                                                                                                          |
| <b>NOS Name</b>            | Minimizing the use of synthetic pesticides and fertilizers and practicing the use of organic and sustainable principles |
| <b>Sector</b>              | Agriculture                                                                                                             |
| <b>Sub-Sector</b>          |                                                                                                                         |
| <b>Occupation</b>          | Post-Harvest                                                                                                            |
| <b>NSQF Level</b>          | 4                                                                                                                       |
| <b>Credits</b>             | 3                                                                                                                       |
| <b>Version</b>             | 1.0                                                                                                                     |
| <b>Last Reviewed Date</b>  | 21/10/2024                                                                                                              |
| <b>Next Review Date</b>    | 21/10/2027                                                                                                              |
| <b>NSQC Clearance Date</b> | 21/10/2024                                                                                                              |

## Qualification Pack

### CUTM/AGR/N0213: Implementation of Seaweed safety standards

#### Description

This NOS focuses on ensuring health and safety in seaweed processing facilities through proactive risk identification, regular hygiene practices, employee training, emergency preparedness, and wellness initiatives. It aims to reduce accidents, occupational illnesses, and promote a healthy work culture by implementing safety standards and monitoring mechanisms for a secure and productive workplace.

#### Scope

The scope covers the following :

- The scope includes implementing health and safety protocols, identifying hazards, ensuring personal and workplace hygiene, conducting emergency response activities, and promoting employee well-being. It applies to all workers involved in seaweed processing and ensures compliance with occupational health and safety standards to maintain a safe and efficient work environment.

#### Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Implementing safety measures like proper signage, training programs, and the provision of personal protective equipment (PPE) helps in preventing injuries, accidents, and occupational hazards.
- PC2.** Monitoring regularly the risk factors which will help in identifying potential hazards associated with seaweed processing activities including trips, slips, cuts, falls, burns, chemical exposures, and ergonomic strains. Strategies are then developed to mitigate these risks effectively.
- PC3.** Providing comprehensive training and education programs ensuring employees are well equipped with the knowledge and skills necessary for performing their duties safely. Training topics may include proper handling of equipment, safety use of chemical, emergency procedures, and ergonomic practices.
- PC4.** Maintaining high standards of regular hygiene and sanitation is to prevent contamination and the spread of foodborne diseases in seaweed processing facilities.
- PC5.** Establishing a vigilant monitoring for any emergency response to enable swift and effective actions to incidents like fires, burns, chemical spills, medical emergencies, and natural disasters.
- PC6.** Implementing health monitoring programs regularly for the detection and prevention of occupational illnesses or injuries among workers.
- PC7.** Promoting employee wellness initiatives, like health and wellness workshops, fitness programs, and mental health support services, fosters a positive work culture and enhances overall employee well-being.

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

## Qualification Pack

- KU1.** Importance of safety measures like signage, PPE, and training in preventing workplace hazards.
- KU2.** Common risk factors in seaweed processing such as chemical exposure, burns, slips, and ergonomic injuries.
- KU3.** Methods for identifying, evaluating, and mitigating workplace hazards.
- KU4.** Significance of regular hygiene and sanitation in preventing foodborne illnesses.
- KU5.** Procedures for handling emergencies like chemical spills, fires, or medical events.
- KU6.** Elements of an effective emergency preparedness plan.
- KU7.** Importance of employee training programs in ensuring safety compliance.
- KU8.** Principles of occupational health monitoring and disease prevention.
- KU9.** Role of mental and physical wellness programs in improving work culture.
- KU10.** Legal and organizational standards related to health, safety, and hygiene at seaweed processing facilities.

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to identify workplace hazards and recommend appropriate safety measures.
- GS2.** Skill in using PPE and promoting its use among co-workers.
- GS3.** Conducting or participating in safety drills and emergency response activities.
- GS4.** Maintaining hygiene standards in workspaces and equipment.
- GS5.** Monitoring and reporting unsafe conditions or incidents immediately.
- GS6.** Supporting training sessions on ergonomics and safe chemical handling.
- GS7.** Assisting in implementing health monitoring and wellness programs.
- GS8.** Communicating safety protocols clearly to team members.
- GS9.** Demonstrating proactive behavior in hazard identification and risk control.
- GS10.** Working as part of a team to create a safe and positive workplace environment.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                                                                                                                                                                                                                                                       | 30           | 70              | -             | -          |
| <b>PC1.</b> Implementing safety measures like proper signage, training programs, and the provision of personal protective equipment (PPE) helps in preventing injuries, accidents, and occupational hazards.                                                                                                          | 4            | 10              | -             | -          |
| <b>PC2.</b> Monitoring regularly the risk factors which will help in identifying potential hazards associated with seaweed processing activities including trips, slips, cuts, falls, burns, chemical exposures, and ergonomic strains. Strategies are then developed to mitigate these risks effectively.            | 4            | 10              | -             | -          |
| <b>PC3.</b> Providing comprehensive training and education programs ensuring employees are well equipped with the knowledge and skills necessary for performing their duties safely. Training topics may include proper handling of equipment, safety use of chemical, emergency procedures, and ergonomic practices. | 4            | 10              | -             | -          |
| <b>PC4.</b> Maintaining high standards of regular hygiene and sanitation is to prevent contamination and the spread of foodborne diseases in seaweed processing facilities.                                                                                                                                           | 5            | 10              | -             | -          |
| <b>PC5.</b> Establishing a vigilant monitoring for any emergency response to enable swift and effective actions to incidents like fires, burns, chemical spills, medical emergencies, and natural disasters.                                                                                                          | 4            | 10              | -             | -          |
| <b>PC6.</b> Implementing health monitoring programs regularly for the detection and prevention of occupational illnesses or injuries among workers.                                                                                                                                                                   | 5            | 10              | -             | -          |
| <b>PC7.</b> Promoting employee wellness initiatives, like health and wellness workshops, fitness programs, and mental health support services, fosters a positive work culture and enhances overall employee well-being.                                                                                              | 4            | 10              | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                                                                                                                                                      | <b>30</b>    | <b>70</b>       | <b>-</b>      | <b>-</b>   |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                            |
|----------------------------|--------------------------------------------|
| <b>NOS Code</b>            | CUTM/AGR/N0213                             |
| <b>NOS Name</b>            | Implementation of Seaweed safety standards |
| <b>Sector</b>              | Agriculture                                |
| <b>Sub-Sector</b>          |                                            |
| <b>Occupation</b>          | Post-Harvest                               |
| <b>NSQF Level</b>          | 4                                          |
| <b>Credits</b>             | 1                                          |
| <b>Version</b>             | 1.0                                        |
| <b>Last Reviewed Date</b>  | 21/10/2024                                 |
| <b>Next Review Date</b>    | 21/10/2027                                 |
| <b>NSQC Clearance Date</b> | 21/10/2024                                 |

## Qualification Pack

### DGT/VSQ/N0102: Employability Skills (60 Hours)

#### Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

#### Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

#### Elements and Performance Criteria

##### *Introduction to Employability Skills*

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

##### *Constitutional values – Citizenship*

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

##### *Becoming a Professional in the 21st Century*

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

##### *Basic English Skills*

To be competent, the user/individual on the job must be able to:



## Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

### *Career Development & Goal Setting*

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

### *Communication Skills*

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

### *Diversity & Inclusion*

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

### *Financial and Legal Literacy*

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

### *Essential Digital Skills*

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

### *Entrepreneurship*

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

### *Customer Service*

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

## Qualification Pack

**PC28.** follow appropriate hygiene and grooming standards

*Getting ready for apprenticeship & Jobs*

To be competent, the user/individual on the job must be able to:

**PC29.** create a professional Curriculum vitae (Résumé)

**PC30.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

**PC31.** apply to identified job openings using offline /online methods as per requirement

**PC32.** answer questions politely, with clarity and confidence, during recruitment and selection

**PC33.** identify apprenticeship opportunities and register for it as per guidelines and requirements

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** need for employability skills and different learning and employability related portals

**KU2.** various constitutional and personal values

**KU3.** different environmentally sustainable practices and their importance

**KU4.** Twenty first (21st) century skills and their importance

**KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

**KU6.** importance of career development and setting long- and short-term goals

**KU7.** about effective communication

**KU8.** POSH Act

**KU9.** Gender sensitivity and inclusivity

**KU10.** different types of financial institutes, products, and services

**KU11.** how to compute income and expenditure

**KU12.** importance of maintaining safety and security in offline and online financial transactions

**KU13.** different legal rights and laws

**KU14.** different types of digital devices and the procedure to operate them safely and securely

**KU15.** how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

**KU16.** how to identify business opportunities

**KU17.** types and needs of customers

**KU18.** how to apply for a job and prepare for an interview

**KU19.** apprenticeship scheme and the process of registering on apprenticeship portal

## Generic Skills (GS)

User/individual on the job needs to know how to:

**GS1.** read and write different types of documents/instructions/correspondence

**GS2.** communicate effectively using appropriate language in formal and informal settings



## Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Introduction to Employability Skills</i>                                                                                                                                                                                                                                                                      | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC1.</b> identify employability skills required for jobs in various industries                                                                                                                                                                                                                                | -            | -               | -             | -          |
| <b>PC2.</b> identify and explore learning and employability portals                                                                                                                                                                                                                                              | -            | -               | -             | -          |
| <i>Constitutional values – Citizenship</i>                                                                                                                                                                                                                                                                       | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC3.</b> recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.                                                               | -            | -               | -             | -          |
| <b>PC4.</b> follow environmentally sustainable practices                                                                                                                                                                                                                                                         | -            | -               | -             | -          |
| <i>Becoming a Professional in the 21st Century</i>                                                                                                                                                                                                                                                               | <b>2</b>     | <b>4</b>        | -             | -          |
| <b>PC5.</b> recognize the significance of 21st Century Skills for employment                                                                                                                                                                                                                                     | -            | -               | -             | -          |
| <b>PC6.</b> practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life | -            | -               | -             | -          |
| <i>Basic English Skills</i>                                                                                                                                                                                                                                                                                      | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC7.</b> use basic English for everyday conversation in different contexts, in person and over the telephone                                                                                                                                                                                                  | -            | -               | -             | -          |
| <b>PC8.</b> read and understand routine information, notes, instructions, mails, letters etc. written in English                                                                                                                                                                                                 | -            | -               | -             | -          |
| <b>PC9.</b> write short messages, notes, letters, e-mails etc. in English                                                                                                                                                                                                                                        | -            | -               | -             | -          |
| <i>Career Development &amp; Goal Setting</i>                                                                                                                                                                                                                                                                     | <b>1</b>     | <b>2</b>        | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> understand the difference between job and career                                                         | -            | -               | -             | -          |
| <b>PC11.</b> prepare a career development plan with short- and long-term goals, based on aptitude                     | -            | -               | -             | -          |
| <i>Communication Skills</i>                                                                                           | <b>2</b>     | <b>2</b>        | -             | -          |
| <b>PC12.</b> follow verbal and non-verbal communication etiquette and active listening techniques in various settings | -            | -               | -             | -          |
| <b>PC13.</b> work collaboratively with others in a team                                                               | -            | -               | -             | -          |
| <i>Diversity &amp; Inclusion</i>                                                                                      | <b>1</b>     | <b>2</b>        | -             | -          |
| <b>PC14.</b> communicate and behave appropriately with all genders and PwD                                            | -            | -               | -             | -          |
| <b>PC15.</b> escalate any issues related to sexual harassment at workplace according to POSH Act                      | -            | -               | -             | -          |
| <i>Financial and Legal Literacy</i>                                                                                   | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC16.</b> select financial institutions, products and services as per requirement                                  | -            | -               | -             | -          |
| <b>PC17.</b> carry out offline and online financial transactions, safely and securely                                 | -            | -               | -             | -          |
| <b>PC18.</b> identify common components of salary and compute income, expenses, taxes, investments etc                | -            | -               | -             | -          |
| <b>PC19.</b> identify relevant rights and laws and use legal aids to fight against legal exploitation                 | -            | -               | -             | -          |
| <i>Essential Digital Skills</i>                                                                                       | <b>3</b>     | <b>4</b>        | -             | -          |
| <b>PC20.</b> operate digital devices and carry out basic internet operations securely and safely                      | -            | -               | -             | -          |
| <b>PC21.</b> use e- mail and social media platforms and virtual collaboration tools to work effectively               | -            | -               | -             | -          |
| <b>PC22.</b> use basic features of word processor, spreadsheets, and presentations                                    | -            | -               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Entrepreneurship</i>                                                                                                                                                          | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC23.</b> identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research                                       | -            | -               | -             | -          |
| <b>PC24.</b> develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion                                                      | -            | -               | -             | -          |
| <b>PC25.</b> identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity                                           | -            | -               | -             | -          |
| <i>Customer Service</i>                                                                                                                                                          | <b>1</b>     | <b>2</b>        | -             | -          |
| <b>PC26.</b> identify different types of customers                                                                                                                               | -            | -               | -             | -          |
| <b>PC27.</b> identify and respond to customer requests and needs in a professional manner.                                                                                       | -            | -               | -             | -          |
| <b>PC28.</b> follow appropriate hygiene and grooming standards                                                                                                                   | -            | -               | -             | -          |
| <i>Getting ready for apprenticeship &amp; Jobs</i>                                                                                                                               | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC29.</b> create a professional Curriculum vitae (Résumé)                                                                                                                     | -            | -               | -             | -          |
| <b>PC30.</b> search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively | -            | -               | -             | -          |
| <b>PC31.</b> apply to identified job openings using offline /online methods as per requirement                                                                                   | -            | -               | -             | -          |
| <b>PC32.</b> answer questions politely, with clarity and confidence, during recruitment and selection                                                                            | -            | -               | -             | -          |
| <b>PC33.</b> identify apprenticeship opportunities and register for it as per guidelines and requirements                                                                        | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                 | <b>20</b>    | <b>30</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                 |
|----------------------------|---------------------------------|
| <b>NOS Code</b>            | DGT/VSQ/N0102                   |
| <b>NOS Name</b>            | Employability Skills (60 Hours) |
| <b>Sector</b>              | Cross Sectoral                  |
| <b>Sub-Sector</b>          | Professional Skills             |
| <b>Occupation</b>          | Employability                   |
| <b>NSQF Level</b>          | 4                               |
| <b>Credits</b>             | 2                               |
| <b>Version</b>             | 1.0                             |
| <b>Last Reviewed Date</b>  | 18/02/2025                      |
| <b>Next Review Date</b>    | 18/02/2028                      |
| <b>NSQC Clearance Date</b> | 18/02/2025                      |

## Assessment Guidelines and Assessment Weightage

### Assessment Guidelines

#### 1. Assessment System Overview:

- Batches assigned to the assessment agencies for conducting the assessment on SIDH or email
- Assessment agencies send the assessment confirmation to TP looping AB/AA to accept in SIDH
- Assessment agency deploys the ToA certified Assessor for executing the assessment
- AB monitors the assessment process & records

#### 2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 30, then there should be 2 Assessors.
- Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.



## Qualification Pack

### 3. Assessment Quality Assurance levels/Framework:

- Question bank is created by the Subject Matter Experts (SME) are verified by the AB
- Questions are mapped to the specified assessment criteria.
- Assessor must be ToA certified & trainer must be ToT Certified

### 4. Types of evidence or evidence-gathering protocol:

- Time-stamped & geotagged reporting of the assessor from assessment location.
- Centre photographs with signboards and scheme specific branding
- Video Call verification/online proctoring will be done during the assessment from AB.

### 5. Method of verification or validation:

- Surprise visit to the assessment location

### 6. Minimum Percentage Required

- 70% overall to pass.
- Minimum 60% in each NOS

7. Reassessment: Candidates failing the assessment will be given one opportunity for reassessment after additional training.

8. Assessor Details: ToA Certified Assessor with relevant experience.

### Minimum Aggregate Passing % at QP Level : 70

**(Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

### Minimum Passing % at NOS Level: 60

**(Please note:** A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a

## Qualification Pack

whole.)

### Assessment Weightage

Compulsory NOS

| National Occupational Standards                                                                                                        | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage  |
|----------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|------------|
| CUTM/AGR/N0201.Seaweed Ecology and its environmental importance                                                                        | 30           | 70              | -             | -          | 100         | 10         |
| CUTM/AGR/N0202.Post-Harvest handling of Seaweed                                                                                        | 30           | 40              | -             | -          | 70          | 20         |
| CUTM/AGR/N0203.Value added seaweed product and preservation                                                                            | 40           | 60              | -             | -          | 100         | 20         |
| CUTM/AGR/N0204.Quality Assurance in seaweed processing                                                                                 | 40           | 60              | -             | -          | 100         | 20         |
| CUTM/AGR/N0205.Minimizing the use of synthetic pesticides and fertilizers and practicing the use of organic and sustainable principles | 30           | 40              | -             | 30         | 100         | 10         |
| CUTM/AGR/N0213.Implementation of Seaweed safety standards                                                                              | 30           | 70              | -             | -          | 100         | 10         |
| DGT/VSQ/N0102.Employability Skills (60 Hours)                                                                                          | 20           | 30              | -             | -          | 50          | 10         |
| <b>Total</b>                                                                                                                           | <b>220</b>   | <b>370</b>      | <b>-</b>      | <b>30</b>  | <b>620</b>  | <b>100</b> |



## Qualification Pack

### Acronyms

|             |                                                 |
|-------------|-------------------------------------------------|
| <b>NOS</b>  | National Occupational Standard(s)               |
| <b>NSQF</b> | National Skills Qualifications Framework        |
| <b>QP</b>   | Qualifications Pack                             |
| <b>TVET</b> | Technical and Vocational Education and Training |

## Qualification Pack

### Glossary

|                                              |                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                | Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.                                                                      |
| <b>Sub-sector</b>                            | Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.                                                                                                                                                                                     |
| <b>Occupation</b>                            | Occupation is a set of job roles, which perform similar/ related set of functions in an industry.                                                                                                                                                                                                |
| <b>Job role</b>                              | Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.                                                                                                                                                                                |
| <b>Occupational Standards (OS)</b>           | OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts. |
| <b>Performance Criteria (PC)</b>             | Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.                                                                                                                                                                    |
| <b>National Occupational Standards (NOS)</b> | NOS are occupational standards which apply uniquely in the Indian context.                                                                                                                                                                                                                       |
| <b>Qualifications Pack (QP)</b>              | QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.                                                                                                                       |
| <b>Unit Code</b>                             | Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'                                                                                                                                                                                                        |
| <b>Unit Title</b>                            | Unit title gives a clear overall statement about what the incumbent should be able to do.                                                                                                                                                                                                        |
| <b>Description</b>                           | Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.                                                                                                                   |
| <b>Scope</b>                                 | Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.                                                                                            |

## Qualification Pack

|                                         |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge and Understanding (KU)</b> | Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.                                                                                                                                               |
| <b>Organisational Context</b>           | Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.                                                                                                                                                                       |
| <b>Technical Knowledge</b>              | Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.                                                                                                                                                                                                                                                               |
| <b>Core Skills/ Generic Skills (GS)</b> | Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles. |
| <b>Electives</b>                        | Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.                                                                            |
| <b>Options</b>                          | Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.                                                                                                                                                          |